Activity Selection Problem

Design and Analysis of Algorithms (DAA) — C Language

Purpose

Select the maximum number of mutually non-overlapping activities using the greedy method. Activities are sorted by increasing finish time before selection.

C Program

```
#include <stdio.h>
#include <stdlib.h>

typedef struct {
    int start;
    int finish;
    int id;
} Activity;

int compare(const void *a, const void *b) {
    const Activity *x = (const Activity *)a;
    const Activity *y = (const Activity *)b;

    if (x->finish < y->finish) return -1;
    if (x->finish > y->finish) return 1;
    return x->start - y->start;
}

void activitySelection(Activity activities[], int n) {
    if (n <= 0) {
        printf("No activities provided.\\n");
        return;
    }

    qsort(activities, n, sizeof(Activity), compare);

    printf("Selected activities (ID: start-finish):\\n");
    int lastFinish = activities[0].finish;
    printf("A%d: %d-%d\\n", activities[0].id,
           activities[0].start, activities[0].finish);

    int count = 1;
    for (int i = 1; i < n; i++) {
        if (activities[i].start >= lastFinish) {
            printf("A%d: %d-%d\\n", activities[i].id,
                   activities[i].start, activities[i].finish);
            lastFinish = activities[i].finish;
            count++;
        }
    }

    printf("Maximum number of non-overlapping activities = %d\\n", count);
}

int main(void) {
    Activity activities[] = {
        {1, 2, 1}, {3, 4, 2}, {0, 6, 3},
        {5, 7, 4}, {8, 9, 5}, {5, 9, 6}
    };
```

```
    int n = sizeof(activities) / sizeof(activities[0]);

    activitySelection(activities, n);
    return 0;
}
```

Sample Output

```
Selected activities (ID: start-finish):
A1: 1-2
A2: 3-4
A4: 5-7
A5: 8-9
Maximum number of non-overlapping activities = 4
```